Computer Vision

T4 Explainable AI

University Master's Degree in
Machine Learning

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Explainable AI

- Introduction
  - CNNs are powerful but opaque → "black boxes"
  - Why it matters:
    - Trust: Ensure the model is making correct decisions.
    - Bias detection: Avoid learning spurious correlations.
    - Regulation: Comply with rules requiring interpretability (e.g., GDPR).
    - Model improvement: Understand failures and generalization.

Explainable AI

- Feature Visualization in Convolutional Layers
  - Provides a window into what each convolutional layer is learning
  - Show that filters are not random and capture hierarchical patterns.
    - Early layers → detect edges, textures, colors.
    - Intermediate layers → capture more complex patterns (shapes, object parts).
    - Deep layers → capture semantic concepts (faces, objects).
  - Feature visualization involves inspecting the feature maps generated by the convolutional filters, giving us insight into which parts of the input image activate specific filters.

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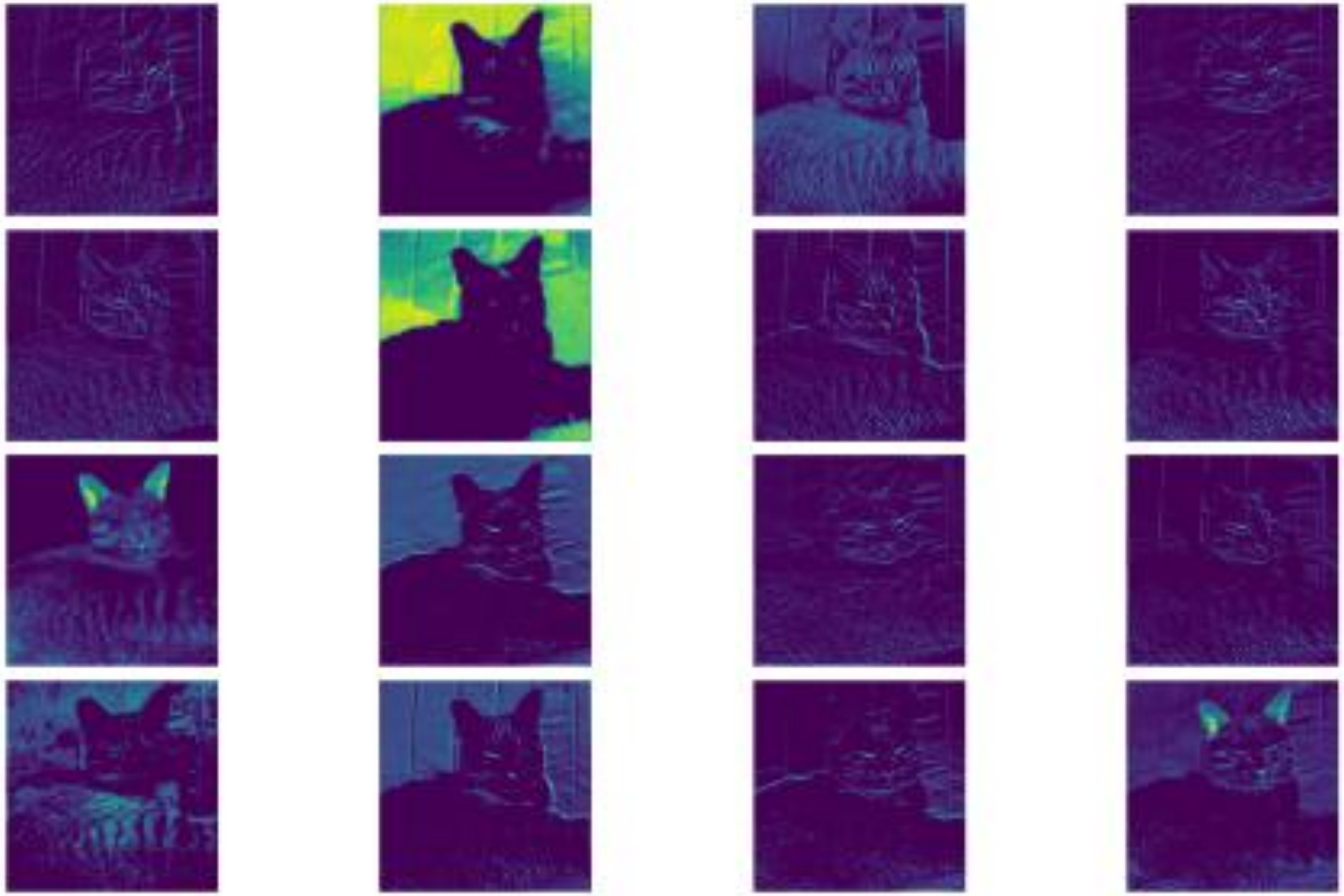

Figure 4.2: Feature Maps from the First Convolutional Layer of VGG16

Explainable AI

- Feature Visualization in Convolutional Layers
  - Challenges in Interpreting Feature Maps
    - Lack of Direct Interpretability
      - Many feature maps detect abstract patterns not easily recognizable by humans.
    - Dependence on Input Data
      - Visualizations vary with different inputs → hard to generalize interpretations.
    - Layer Complexity
      - Deeper layers produce more complex feature maps.
      - Advanced techniques (e.g., activation maximization, saliency maps) may be needed to understand them.
  - To gain deeper insights, more sophisticated interpretability methods are needed

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- Feature Visualization interesting links
  - https://distill.pub/2017/feature-visualization/
  - https://distill.pub/2018/building-blocks/
  - https://distill.pub/2020/circuits/early-vision/
  - https://distill.pub/2019/activation-atlas/

Explainable AI

- Feature attribution methods
  - Saliency maps
  - Integrated Gradients
  - Grad-CAM

Explainable AI

- Feature attribution methods
  - Saliency maps (Simonyan et al., 2014)
    - Gradient of the class w.r.t. the input → highlights relevant pixels.

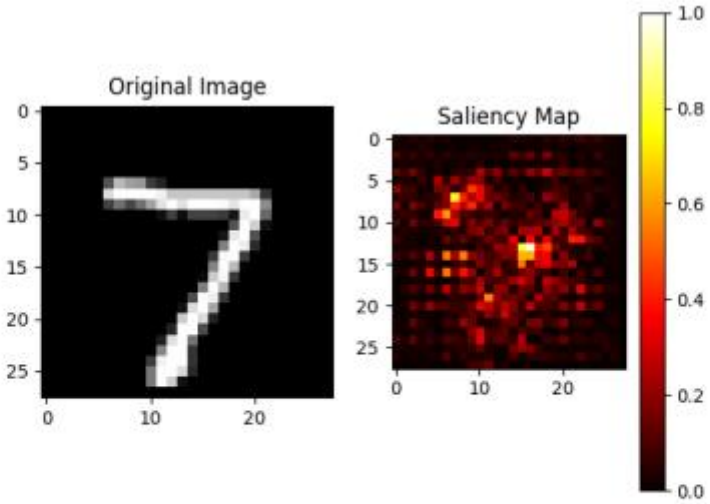

Figure 6.6: Saliency map for an MNIST image. The map shows which pixels the model focused on when predicting the digit, with warmer colors indicating higher importance.

Explainable AI

- Feature attribution methods
  - Saliency maps (Simonyan et al., 2014)
    - Limitations
      - Gradient Saturation: If the model's output is saturated, the gradient values may be close to zero, making the saliency map less informative.
      - Noise and Artifacts: Saliency maps can be noisy and may highlight irrelevant features, especially in high-dimensional inputs.

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- Feature attribution methods
  - Integrated Gradients
    - Integrates gradients from a reference (e.g., black image) to the actual image.
      - Compute the accumulated gradients of the model's output with respect to the input features as the input transitions from a baseline (e.g., a zero vector or an all-blank input) to the actual input.
      - The attribution score for each feature is obtained by summing these gradients along the integration path.
    - Reduces saturation and noise problems of saliency maps.

Explainable AI

- Feature attribution methods
  - Integrated Gradients

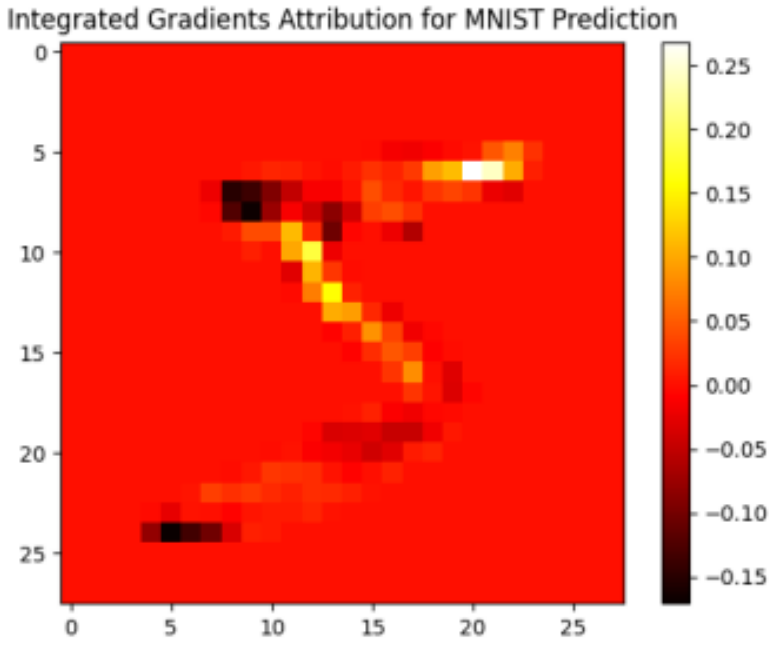

Figure 6.4: Integrated Gradients attribution map for a sample MNIST image. The highlighted pixels reveal the model's focus areas when classifying the digit.

Explainable AI

- Feature attribution methods
  - Integrated Gradients
    - Advantages:
      - Theoretical Soundness: Satisfies properties like completeness — attributions sum to the difference between model output for input and baseline.
      - Model-Agnostic: Applicable to any differentiable model (CNNs, LLMs, etc.).
      - No Gradient Saturation: Integrates gradients over a range of inputs, reducing saturation issues.
    - Limitations:
      - Choice of Baseline: Results depend on baseline; wrong choice may mislead.
      - Computational Cost: Requires multiple model evaluations → expensive for large models or high-dimensional data.

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- Feature attribution methods
  - Grad-CAM (Gradient-weighted Class Activation Mapping)
    - Uses gradients of the target class w.r.t. a convolutional layer to weight activation maps.
    - Produces a heatmap over the original image → easy to interpret.
    - Works even with very deep models and preserves spatial structure.

Explainable AI

- Feature attribution methods
  - Grad-CAM (Gradient-weighted Class Activation Mapping)
    - Uses gradients of the target class w.r.t. a convolutional layer to weight activation maps.
    - This gradient information is used to weight the feature maps, highlighting the most important regions for the prediction.
    - Produces a heatmap over the original image → easy to interpret.
    - Works even with very deep models and preserves spatial structure.
  - Grad-CAM++ refines this approach by considering higher-order derivatives, improving localization and providing better explanations for multi-object scenarios

Explainable AI

- Feature attribution methods
  - Grad-CAM (Gradient-weighted Class Activation Mapping)

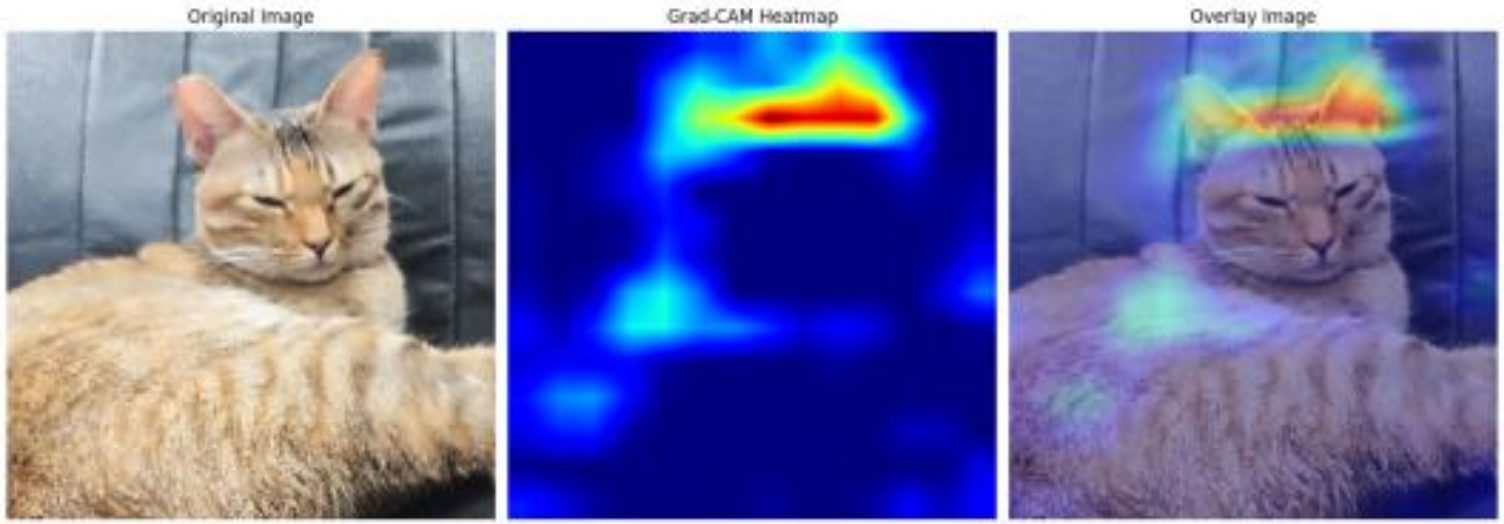

Figure 6.8: Example of Grad-CAM on a Pre-trained VGG16 Model: Original Image, Grad-CAM Heatmap, and Overlay Image

Explainable AI

- Feature attribution methods
  - Grad-CAM (Gradient-weighted Class Activation Mapping)
    - Advantages:
      - Visual Interpretability: Heatmaps are intuitive, highlighting important input regions.
      - Model-Agnostic: Works with any model having convolutional layers.
      - Improved Localization (Grad-CAM++): Uses higher-order gradients to better localize important features.
    - Limitations:
      - Dependence on Convolutional Layers: Not directly applicable to models without convolutions (e.g., some NLP models).
      - Coarse Heatmaps: May lack precise localization in complex, multi-object scenes.

Explainable AI

- Summary
  - Explainable AI opens the “black box” of CNNs and helps:
    - Understand learned patterns.
    - Assess model reliability.
    - Communicate results to non-experts.
  - Gradient-based methods focus on input attribution, activation-based methods focus on internal layer activations.

Method	What it shows	Advantages	Limitations
Feature Visualization	Activations of intermediate CNN layers	Reveals how features evolve (edges → textures → objects)	Hard to interpret in deeper layers; abstract patterns
Saliency Maps	Pixel-level gradients of output w.r.t input	Simple, fast, intuitive; highlights sensitive pixels	Often noisy; lacks spatial coherence
Integrated Gradients	Pixel-level contributions via baseline-to-input path	Theoretically sound (completeness), reduces noise and saturation	Sensitive to baseline choice; computationally expensive
Grad-CAM	Class-specific heatmaps over input regions	Intuitive, localized, easy to overlay on images	Requires conv layers; heatmaps can be coarse